

Improved dissimilarity-cluster IV

Question and estimand

Estimate the elasticity of H-2A demand with respect to the AEWR. The design uses wage innovations in distant, economically disconnected parts of the same AEWR region to predict the region-wide AEWR faced by a target cluster.

The estimand is local to AEWR movements predicted by the retained donor wage shocks.

Main changes to the current design

Current design	Revised design
Five clusters and two donors in every region	Region-specific constrained clusters
No minimum cluster size or mass	At least 3 CZ-region units and 5% of baseline hired-worker hours
Donors selected only by agro-dissimilarity	Donors must also be geographically and economically disconnected
Annual weights calibrated to the realized FLS wage	Frozen pre-period weights; no FLS or AEWR calibration
Annually updated frame	Fixed baseline worker-hour frame
County-year estimation	Target-cluster-year estimation
Region-by-cluster standard errors	AEWR-region inference with few-cluster and weak-IV procedures

Cluster and donor construction

1. Define units as CZ-by-AEWR-region intersections.
2. Use only pre-sample crop mix, harvest timing, climate, soil, and farm structure.
3. Construct an exclusion graph. A target-donor pair is inadmissible if it:
 - lies within a pre-specified distance;
 - shares important employers, farm labor contractors, or associations;
 - has substantial crop-by-harvest-calendar overlap; or
 - has strong pre-period residual wage or H-2A co-movement.
4. Form clusters subject to minimum unit and baseline worker-hour mass constraints. Let the number of clusters vary by region.
5. Choose donors only from admissible clusters. Require an effective donor-area count of at least five and cap any area's weight at 30%.
6. Freeze the cluster map, donor map, and weights before examining the first or second stage.

Exclude a region if these constraints are infeasible; do not relax them after examining results. Cluster stability under alternative feature weights and small

data perturbations is a required diagnostic.

Instrument

For target cluster g in region r , define

$$Z_{grt} = \sum_{a \in D_{gr}} s_{gra}^0 \log w_{a,t-1}^{OEWs},$$

where D_{gr} is the fixed admissible donor set and s_{gra}^0 is a frozen baseline hired-worker-hour share.

OEWS is an external predictor of the FLS wage, not a mechanical component of it. The primary instrument must not use realized FLS wages, AEWRs, quarterly FLS targets, H-2A outcomes, future Census benchmarks, or annual endogenous frame updates.

Use ΔZ_{grt} to instrument $\Delta \log AEWR_{rt}$ in a first-difference robustness specification.

Estimation

$$\log Y_{grt} = \alpha_g + \lambda_t + \beta \log AEWR_{rt} + X'_{grt} \gamma + \varepsilon_{grt},$$

instrumenting $\log AEWR_{rt}$ with Z_{grt} . The primary outcome is requested positions or applications; certified positions and contract hours are secondary. Use a fixed baseline farm-employment offset where appropriate.

Estimate at the target-cluster-year level. Cluster at the AEWR-region level and report CR2 or restricted wild-cluster-bootstrap inference together with Anderson–Rubin or CLR confidence sets.

Identifying assumption

Conditional on fixed effects and controls, retained donor wage innovations affect target-cluster H-2A demand only through the AEWR. Agro-dissimilarity alone is insufficient; the exclusion graph is intended to remove the main geographic, commodity, employer-network, and worker-mobility channels.

Required diagnostics

- Region-clustered first stage and weak-IV-robust confidence sets.
- Instrument concentration and effective donor count.
- Leads of the instrument and pre-period outcome trends.
- Leave-one-region, donor-cluster, and OEWS-area-out estimates.
- Separate distant-donor and crop-dissimilar-donor instruments.
- Results excluding all shared employer/FLC networks.

- Results for 2011–2019 separately from 2020–2022.
- Permutations over admissible donor maps and plausible-exogeneity bounds.

The design should remain supporting evidence unless these diagnostics show both a strong first stage and limited sensitivity to donor composition.